

HEEYEON KOO

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SUMMARY

University of Texas, Dallas

Master of Data Science and Statistics/Concentration: Applied Statistics

Richardson, Texas

Aug. 2025 – Present

Yonsei University

Master of Artificial Intelligence (GPA: 4.15/4.5)

Seoul, Korea

Sept. 2023 – Aug. 2025

- Teaching Assistant, “Computer Programming” (Fall 2024)
- Teaching Assistant, “Ontology Engineering” (Fall 2024)
- Admission Scholarship (Fall 2023)

Kyung Hee University

Bachelor of Computer Science and Engineering (GPA: 3.91/4.5)

Gyeonggi-Do, Korea

Mar. 2018 – Feb. 2023

- Academic Excellence Scholarship (Spring 2018); Software Scholarship (Spring 2022)

PUBLICATIONS

Koo, H.Y., Kim, D.H., Choi, H.J., & Lee, K.H. (2025) “FLOW: Complex Fact-Checking through Linking Implicit and Explicit Claims with Large Language Model Reasoning”, Under review.

Shim, M.D., Choi, H.J., **Koo, H.Y.**, & Um, K.H. “OmEGa (Ω): Ontology-based Information Extraction Framework for Constructing Task-centric Knowledge Graph from Manufacturing Documents with Large Language Model,” *Advanced Engineering Informatics (ADVEI)*, Accepted.

Choi, W. S., Lee, H., Han, D. S., Park, J., **Koo, H.**, & Zhang, B. T. (2024, Mar.). “DUEL: Duplicate Elimination on Active Memory for Self-Supervised Class-Imbalanced Learning.” In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 38, No. 10, pp. 11579-11587), Accepted.

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, Java, MATLAB

Libraries: PyTorch, scikit-learn, Spacy, HuggingFace, Transformer, Pandas, NumPy

Web & Mobile Development: HTML, CSS, React, NodeJS, MySQL, Android

EXPERIENCE

Internet Computing Laboratory | Yonsei University

Seoul, Korea

Graduate Research Student

Aug. 2023 – Present

- Collaborate on a corporate project with Hyundai to develop an ontology-based information extraction framework, aimed to improve data retrieval efficiency from large-scale datasets.
- Establish annotation guidelines for document data to facilitate Named Entity Recognition (NER) tasks, improving model accuracy; perform fine-tuning interactions approximately 10 times and evaluate the model's performance.

Bio-Intelligence Laboratory | Seoul National University

Seoul, Korea

Research Intern

Jan. 2023 – Apr. 2023

- Contributed to a paper on bioinformatics and AI applications for publication, assisting in drafting sections on methodology and results interpretation using LaTeX.

Electronics and Telecommunications Research Institute

Daejeon, Korea

Research Intern, Media Research Division

Jul. 2022 – Aug. 2022

- Analyzed the code of interference codecs, focusing on hologram data compression; identified key performance bottlenecks and proposed solutions to optimize data transmission efficiency.
- Prepared technical documents and presentation materials for internal meetings, explaining codec performance metrics and potential applications for hologram compression.
- Simulated JPEG, HEVC, and interference codecs using hologram data, analyzing compression quality and speed, and comparing results with industry-standard benchmarks.

Visual Media Laboratory | Kyung Hee University*Undergraduate Research Student***Gyeonggi-do, Korea***Aug. 2021 – Dec. 2022*

- Researched and simulated video codecs such as HEVC and VVC using Python and MATLAB to assess performance and efficiency; developed custom simulations to compare compression techniques.
- Participated in a 'Deep Learning Seminar' during winter break, training various neural networks on real-world datasets focusing on optimization techniques and model accuracy.
- Attended graduate-level lecture 'Digital Holography,' reviewing its applications and challenges in modern media.

Software Engineering Intern | Myzzym**Gyeonggi-do, Korea***Dec. 2020 – Feb. 2021*

- Developed a storage platform using React.js, ensuring a responsive, user-friendly interface; worked with Redux for state management and integrated backend APIs for seamless data handling.
- Created test scenarios for user and admin modes to ensure functionality across varying access levels; led testing processes, addressing key bugs to improve platform performance; studied object detection algorithms using YOLO.

PROJECT ACTIVITIES**Software Scholarship Program | Kyung Hee University**

Spring 2022

- Organized a team and successfully conducted regular group learning activities, such as creating Python content, through platforms like Google Drive and Band.
- Monitored student attendance and issued certificates based on their participation.

Algorithm Study | Kyung Hee University 'D.COM' Club

Fall 2020

- Solved various algorithm problems on Python, discussion solutions with club members on Zoom.

Independent Projects

Spring 2020

- Developed the "Fish-Bread Game" using Java, where players bake fish-shaped bread over time and earn points.
- Created a project that displays movie rankings and additional details.

Android Study | Kyung Hee University 'D.COM' Club

Spring 2022

- Participated in presentations on clean code, fragment, recycler view, etc.
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